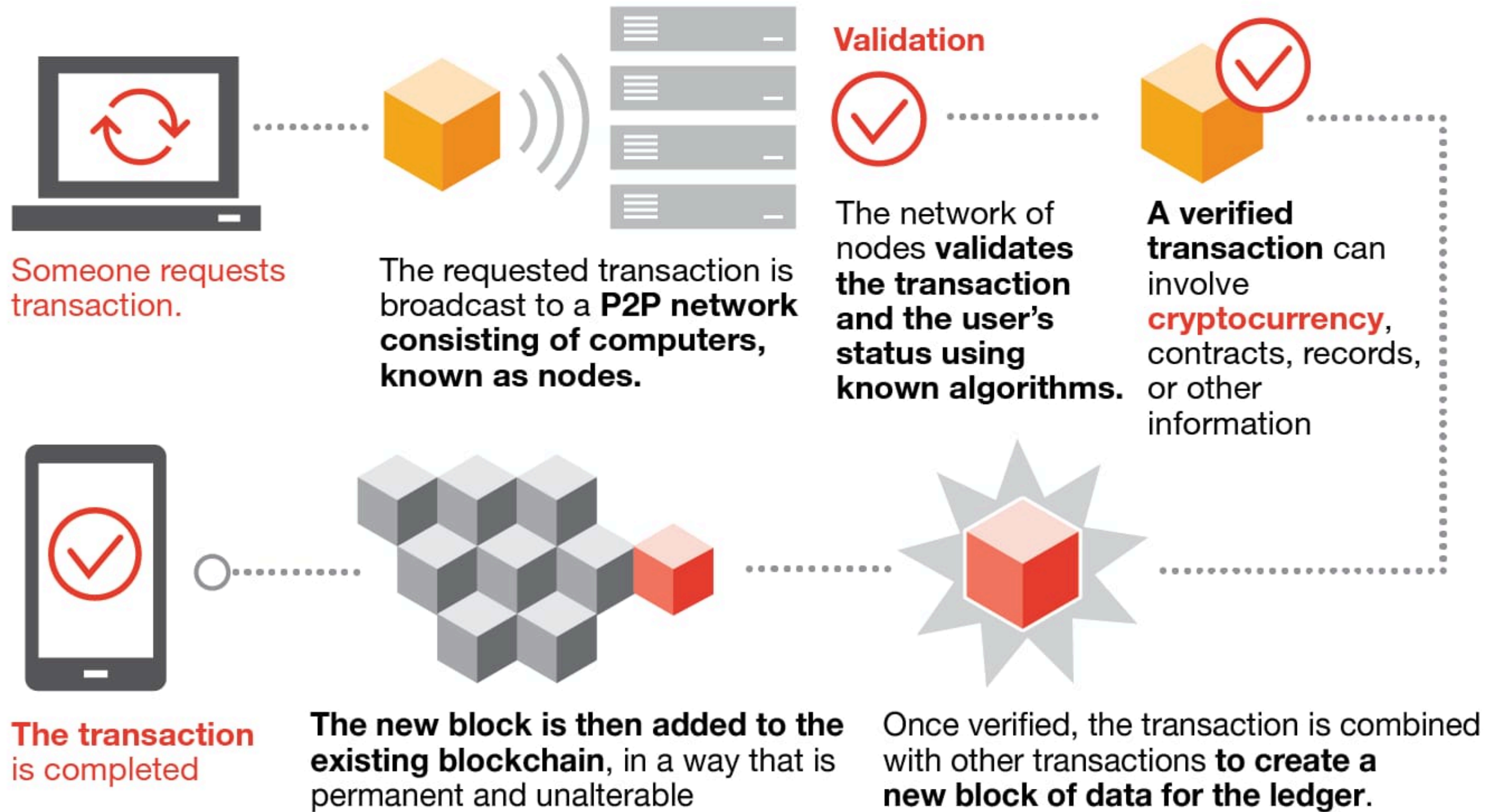


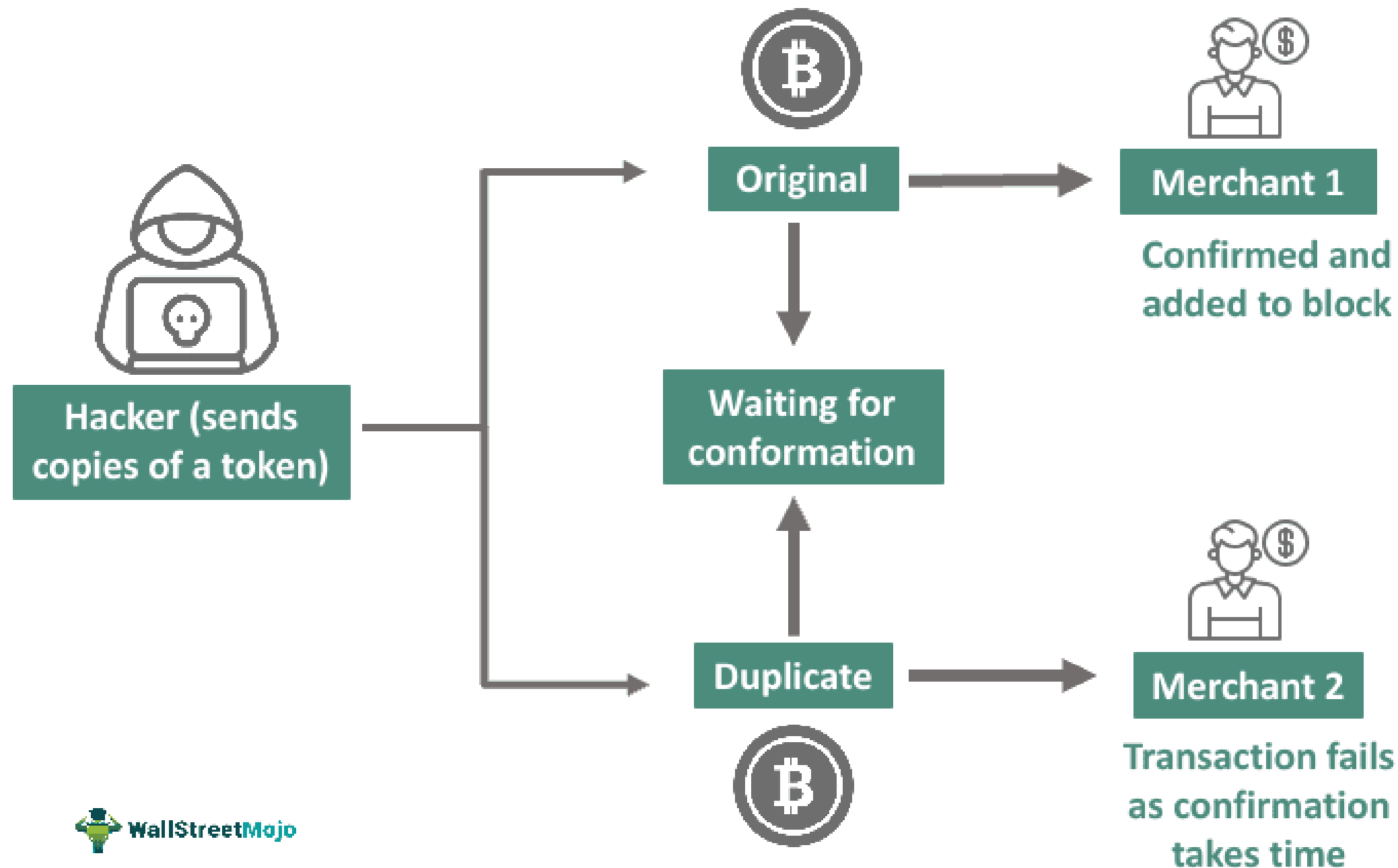
The Double-Spending Problem. Could It Still Threaten Cryptocurrencies?

YES

How Transactions work?

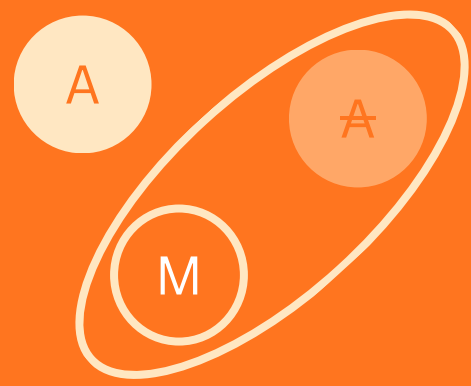


What Is Double-Spending?

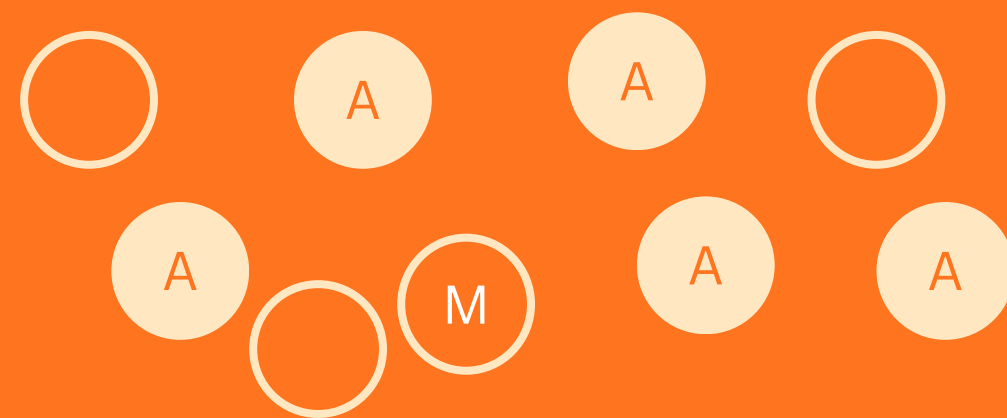


Double-spending is spending the same cryptocurrency or blockchain token more than once.

Double-spending is a significant threat to blockchain networks, potentially allowing a token or cryptocurrency to be used more than once, undermining trust in digital transactions.

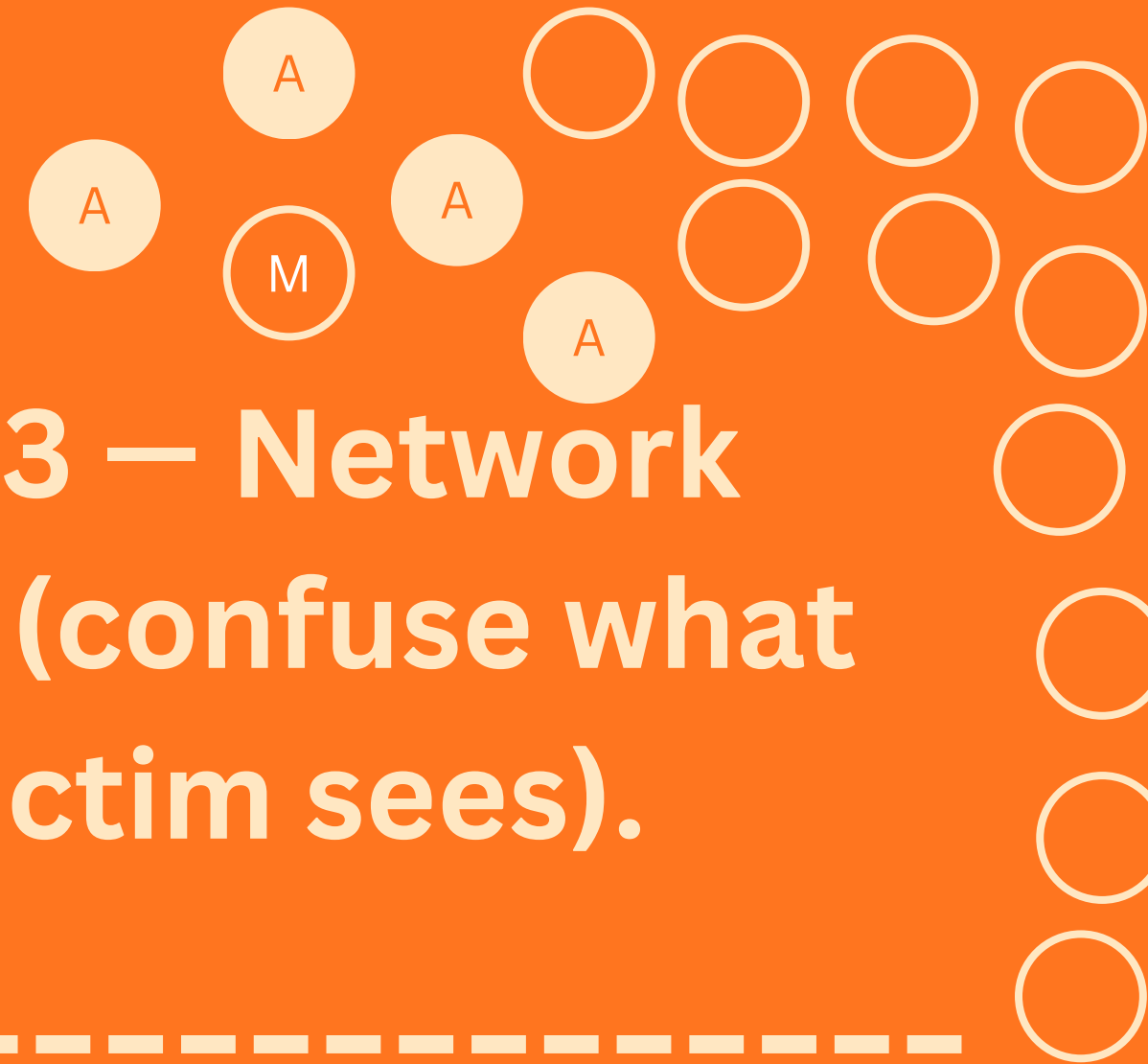


Family 1 — Pre-confirmation tricks (the ‘catch you before it settles’ attacks).



Family 2 — Chain reorganization power (rewrite recent history).

Related Attacks



Family 3 — Network isolation (confuse what the victim sees).

F1:

- Race / zero-confirmation double-spend.
- Finney attack.

F2:

- 51% attack
- Selfish Mining

F3:

- Sybil attacks
- Eclipse-style attacks

How To Prevent?

- Enhanced consensus mechanism

-
- Block timestamps

-
- Transaction copies

-
- Inclusion of nonce

-
- Confirmation of Unspent Transaction Output (UTXO)

So... Does double-spending still threaten cryptocurrencies?

- **Ethereum Classic (ETC)** — Jan-2019: $\approx 219,500$ ETC double-spent ($\sim \$1.1\text{M}$). Jul–Aug-2020: $\approx 807\text{k}$ ETC and $\approx 458\text{k}$ ETC double-spent across multiple 51% attacks (multi-thousand-block reorgs).
- **Bitcoin Gold (BTG)** — May-2018: $\approx \$18\text{M}$ double-spent; additional incidents reported in 2020.
- **Horizen/ZenCash** — Jun-2018: $\approx \$550\text{k}$ double-spent.
- **Monaco** — May-2018: deep reorg event enabling double-spend attempts.

References

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Thanks!!!!!!